

Acme Analytics Corp.

Incident Report — System Outage, October 14, 2024

Classification: Internal | Prepared by: DevOps Team

1. Executive Summary

On October 14, 2024 at 02:31 UTC, the Cloud Analytics Pro production environment experienced an unplanned outage lasting 47 minutes. Approximately 1,240 active users were affected. Root cause was a misconfigured Kubernetes rolling-update policy that caused all pods to restart simultaneously, exhausting the connection pool to the primary PostgreSQL cluster.

2. Timeline of Events

- 02:31 UTC — Monitoring alerts fire: API error rate exceeds 40% threshold.
- 02:33 UTC — On-call engineer (M. Chen) acknowledges PagerDuty alert.
- 02:38 UTC — Root cause identified: maxUnavailable=100% in prod manifest.
- 02:41 UTC — Rollback initiated to previous stable image (v2.14.3).
- 03:02 UTC — 90% of pods healthy; error rate returns below 1%.
- 03:18 UTC — Full recovery confirmed. Post-mortem scheduled.

3. Impact Assessment

- Affected users : 1,240 (all tiers)
- Revenue impact : ~\$4,200 estimated SLA credit exposure
- Data integrity : No data loss confirmed. All writes queued and replayed.
- SLA breach : Yes — Enterprise tier SLA (99.9% monthly) breached.

4. Root Cause Analysis

The outage was caused by `maxUnavailable: 100%` in the Kubernetes deployment manifest introduced in PR #4821. This setting was accidentally merged into the production branch during a late-night hotfix push on October 13.

Contributing factors

- a) Missing branch protection on `k8s/prod/` manifest directory.
No CODEOWNERS review gate & single engineer approval was sufficient.
- b) Staging and production manifests in same repo with similar directory names (`staging/` vs `prod/`). Engineers under time pressure misidentify.
- c) Pre-deployment validation pipeline lacked a policy check for `maxUnavailable` values above 30% in production namespaces.
- d) Connection pool exhaustion amplified blast radius. PostgreSQL primary configured with `max_connections=200`; analytics-api holds 8 connections per pod x 22 pods = 176 connections. Simultaneous restarts caused a thundering-herd reconnection storm consuming all available connections.

5. Detection and Response

Automated monitoring detected the anomaly within 2 minutes via Datadog synthetic checks and PagerDuty escalation. On-call runbook directed the engineer to check deployment history first, shortening MTTR significantly.

MTTR 47 minutes | MTTD 2 minutes | MTTR target 30 minutes

6. Remediation Actions and Owners

All items approved by post-mortem review board, October 17, 2024.

Action Item	Owner	Priority	Due Date	Status
Add CODEOWNERS to k8s/prod/	Platform Eng	P0	Oct 21	Done
Policy: maxUnavailable <= 30% prod	DevOps	P0	Oct 24	Done
Rename staging/ to stg/	DevOps	P1	Nov 01	In Progress
Increase PostgreSQL max_connections=400	DBA Team	P1	Nov 08	Planned
PgBouncer in front of primary DB	Platform Eng	P1	Nov 15	Planned
Add pre-deploy diff tool to CI	DevOps	P2	Nov 22	Planned
Chaos drill: simultaneous pod restart	SRE	P2	Dec 06	Planned
Runbook: connection pool exhaustion	On-Call Lead	P2	Nov 01	In Progress

7. Long-term Preventive Investment (Q1 2025)

Engineering leadership has approved the following investments:

- GitOps operator (ArgoCD) with environment-aware policy enforcement replacing manual kubectl apply workflows across all environments
- Service mesh (Istio) for circuit-breaking and graceful degradation during downstream failures
- Centralised secrets and config management (HashiCorp Vault) to eliminate environment config bleed between staging and production.